



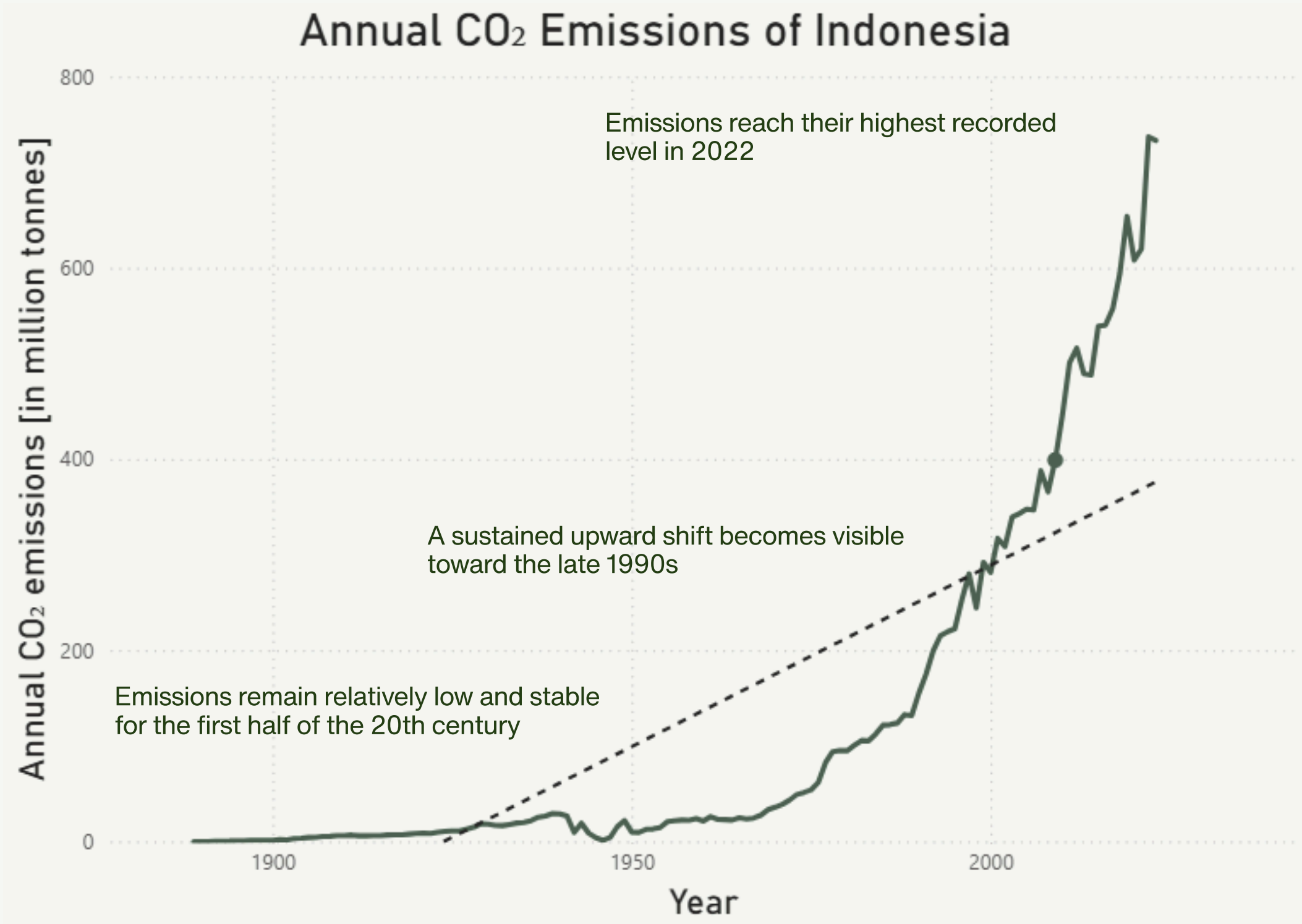
INDONESIA: THE PIVOT POINT FOR SOUTHEAST ASIA'S CO₂



Not the region's highest per-capita emitter, but the largest absolute driver due to scale, growth, and coal reliance.

TREND OF INDONESIA'S CO₂ EMISSIONS

Indonesia's CO₂ emissions show a clear structural upward shift beginning in the late 20th century, followed by rapid acceleration in the 21st century.



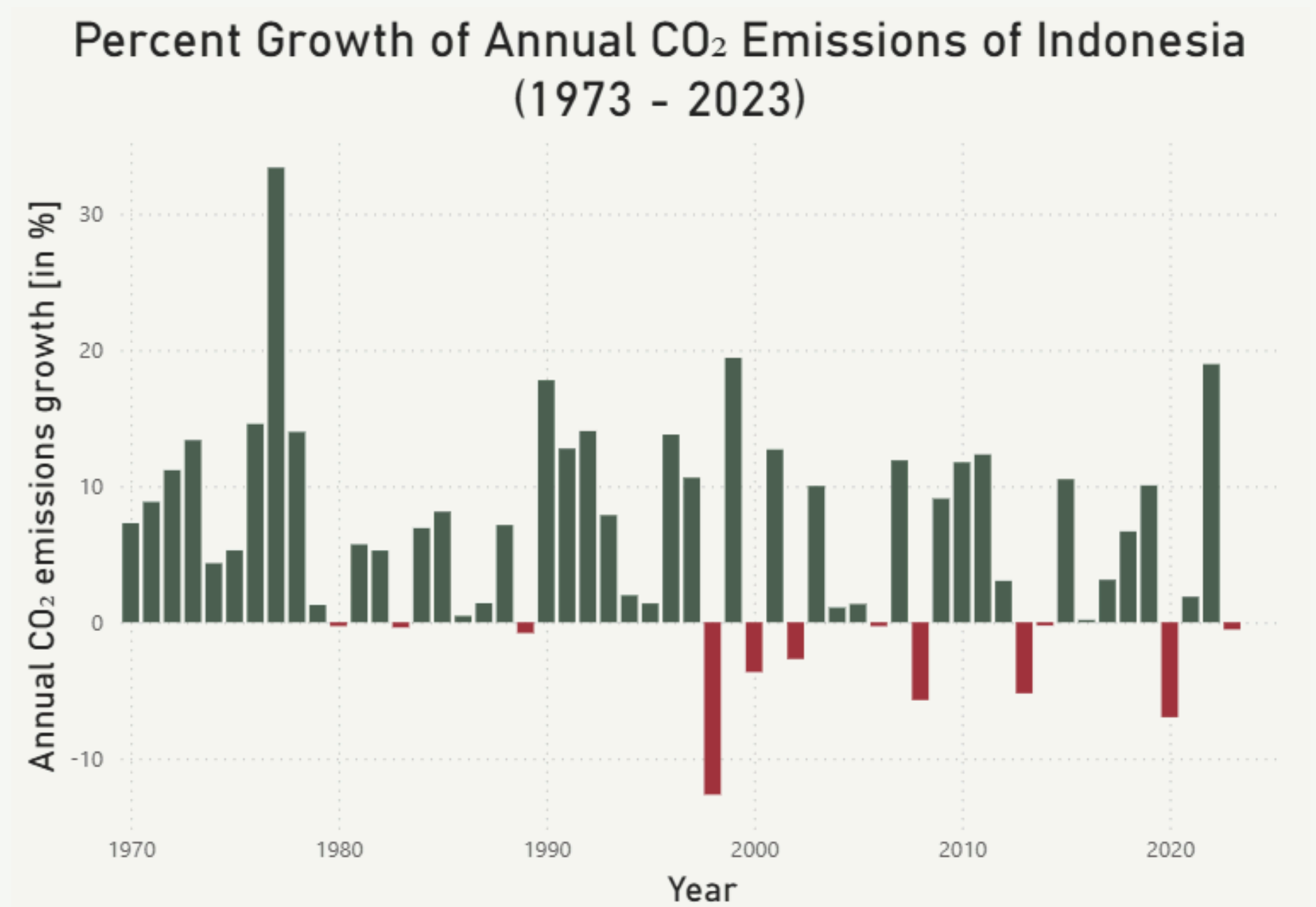
Maximum CO₂
(Mt)
737.07

Year: 2022

YEAR-ON-YEAR GROWTH IN INDONESIA'S CO₂ EMISSIONS

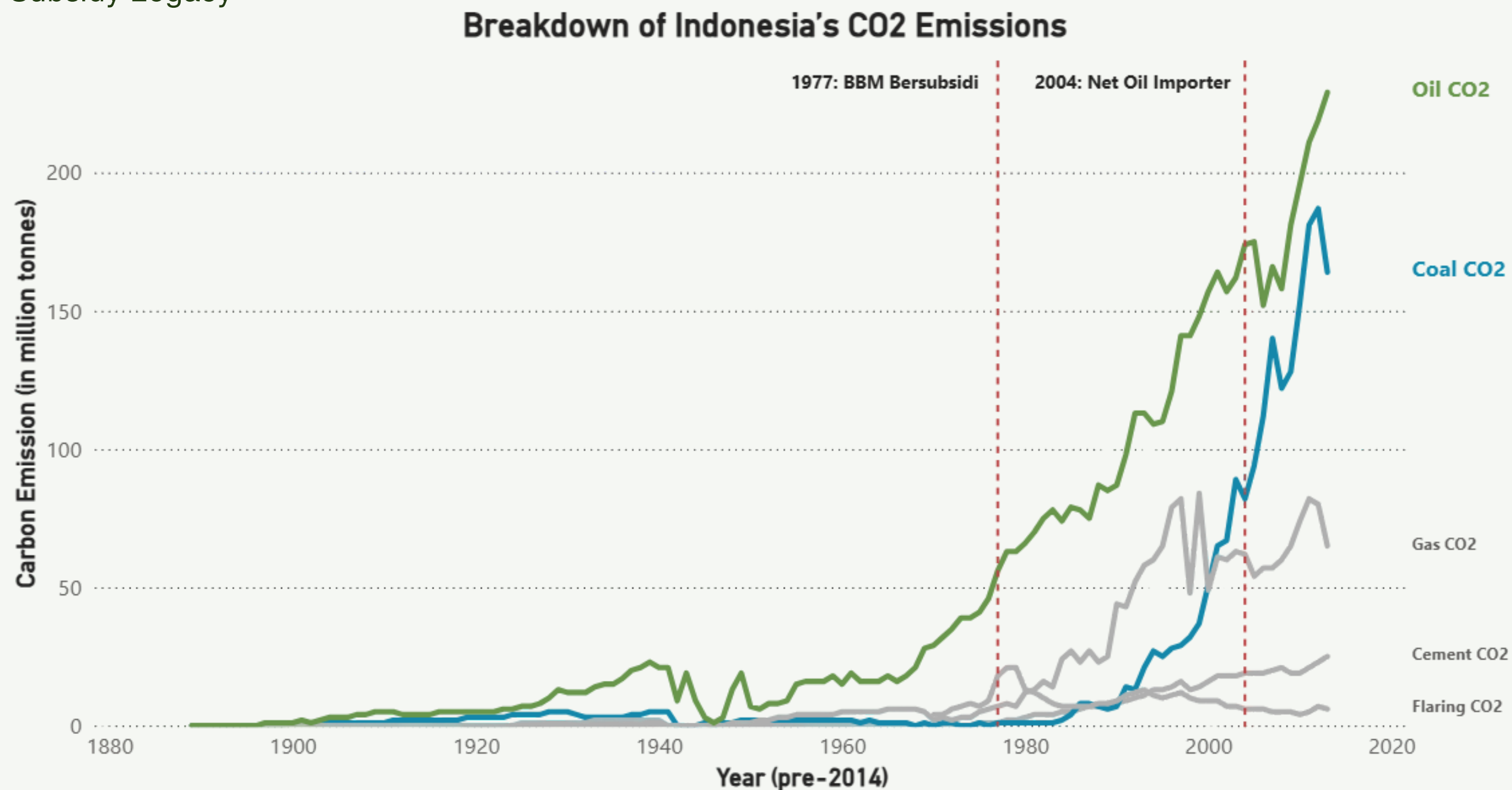
Despite volatility, year-on-year emissions growth has remained largely positive over the past five decades.

- Most years show positive growth rather than contraction
- Sharp dips are limited and often followed by rebounds
- Volatility increases in the 21st century as total emissions grow larger



HISTORICAL BASELINE: INDONESIA'S OIL-DRIVEN GROWTH

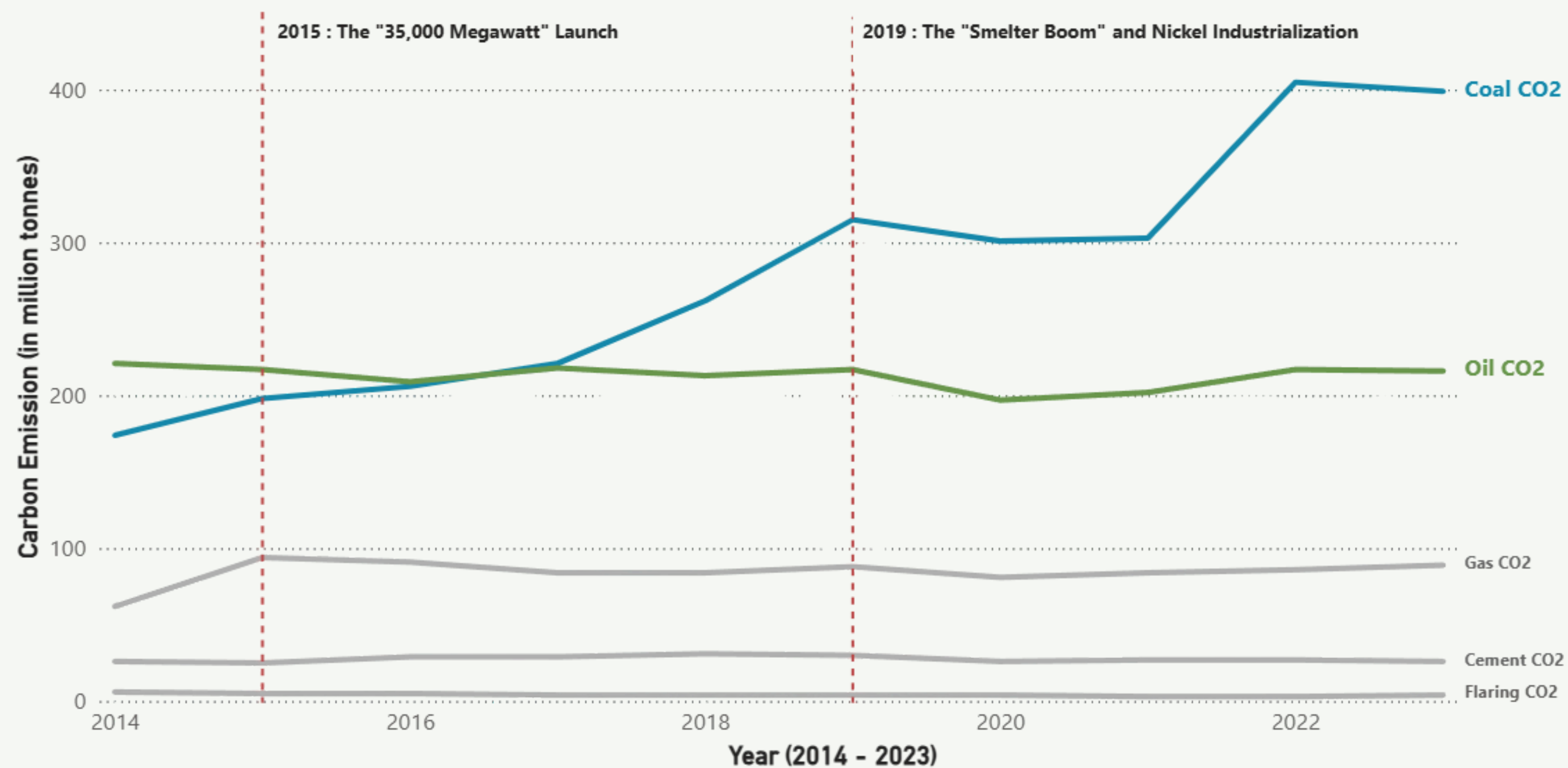
The Oil Era & The Subsidy Legacy



INDUSTRIAL POWER & GRID LOCK-IN

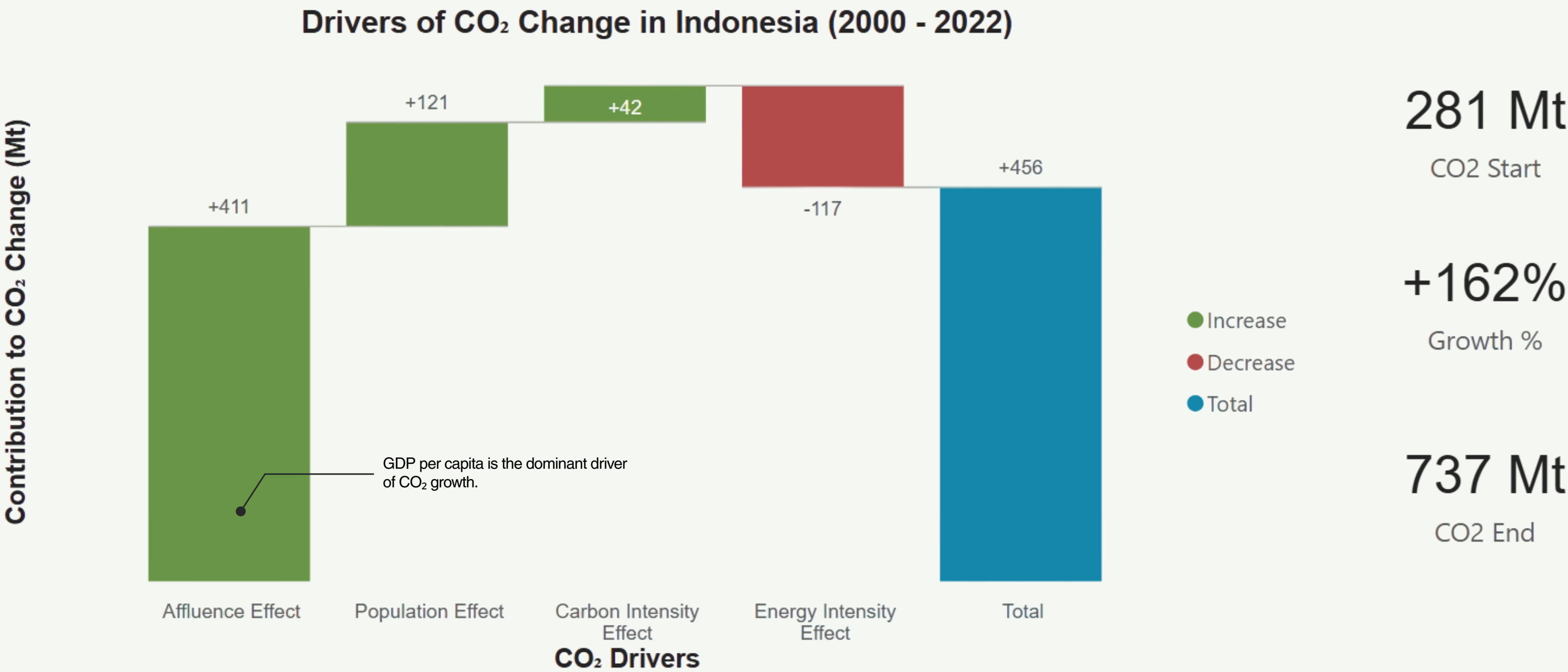
The shift to coal-heavy baseload power and industrial captive plants.

Breakdown of Indonesia's CO2 Emissions



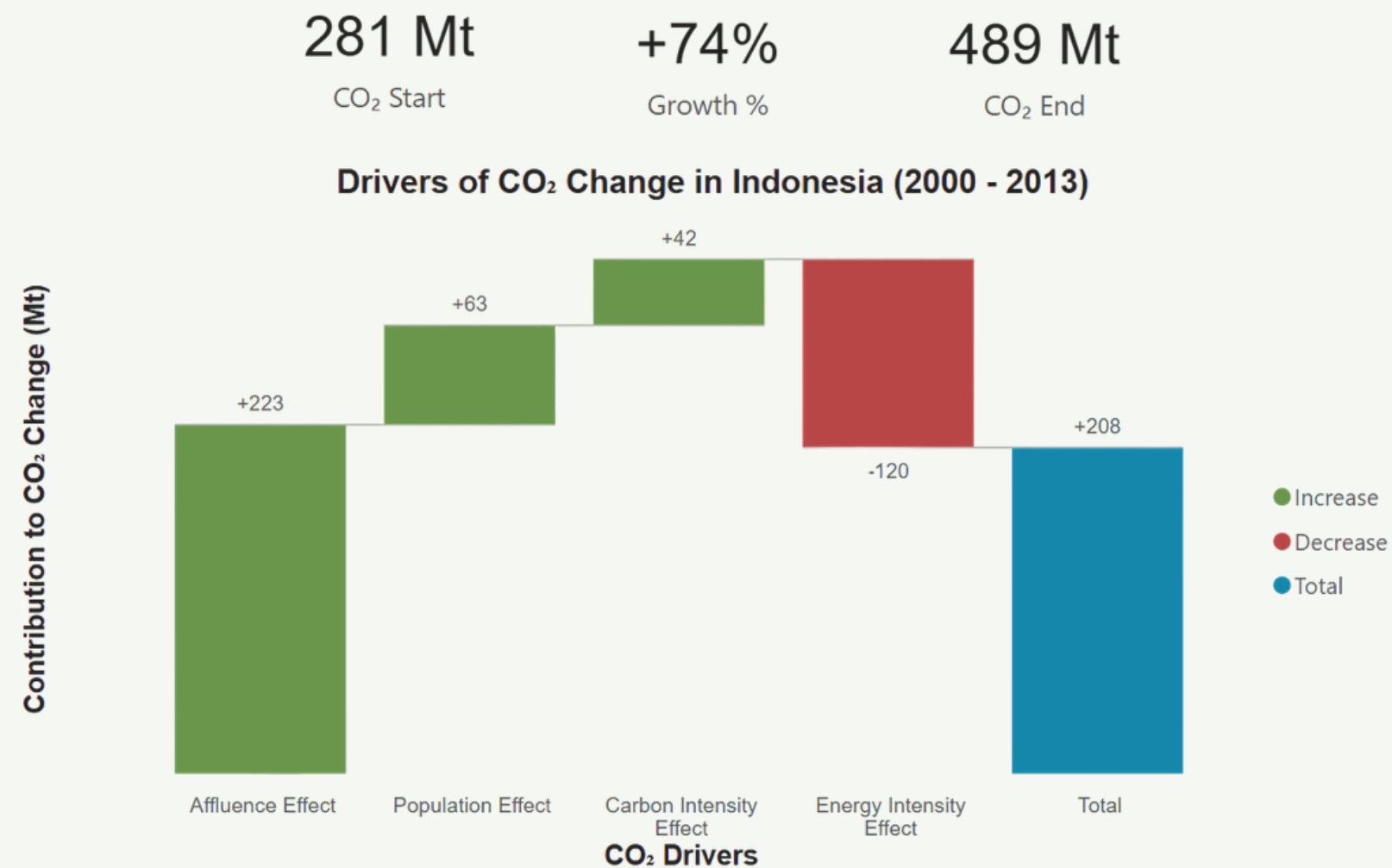
WHAT'S DRIVING THE RISE OF EMISSIONS?

Waterfall decomposition of total CO₂ change (2000–2022)



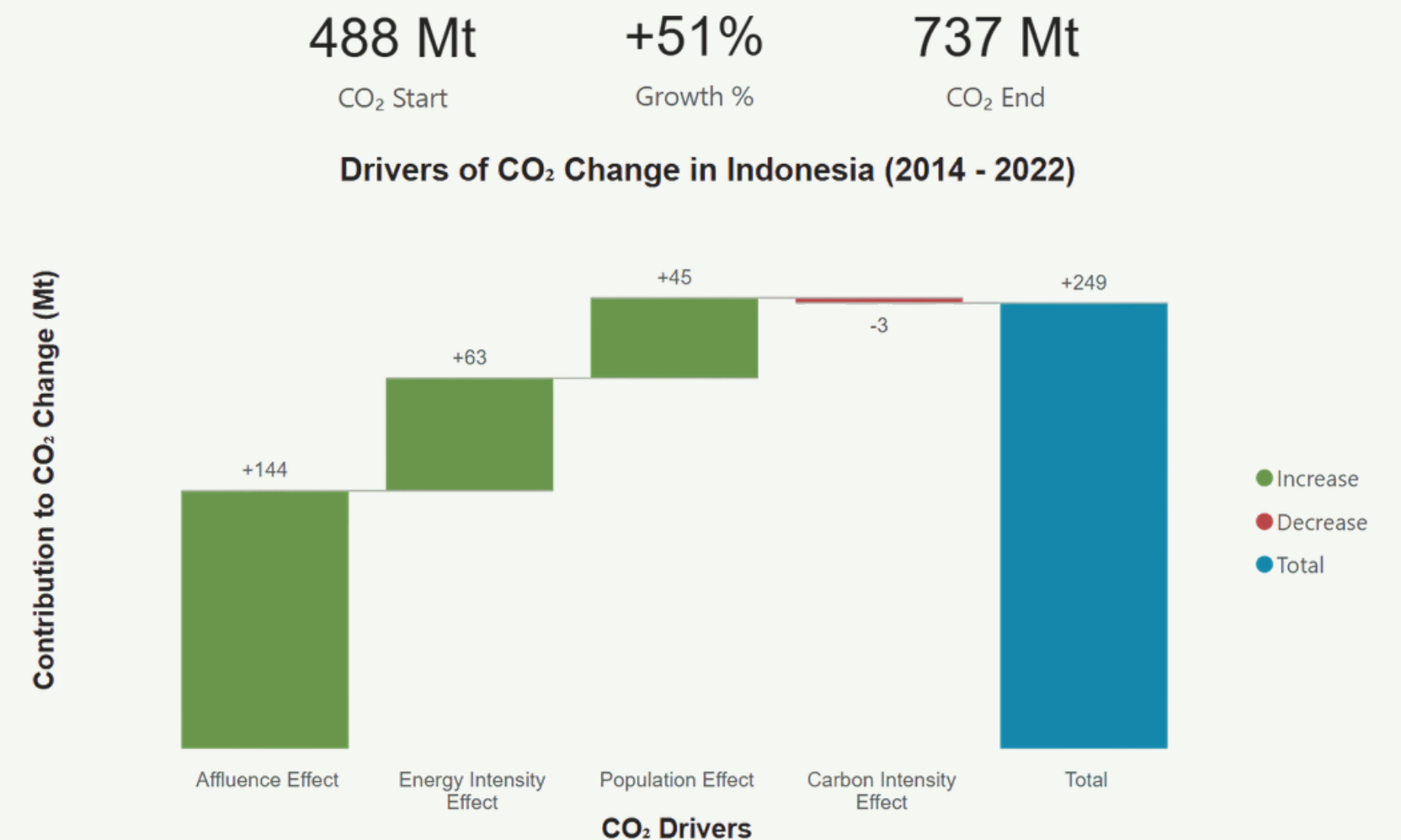
THE "POLICY PARADOX" (PRE VS. POST 2014)

How the 2015 Coal Push Stalled Indonesia's Efficiency



Decoupling Success

Before 2014, efficiency saved 120 Mt of CO₂. We were growing smarter.

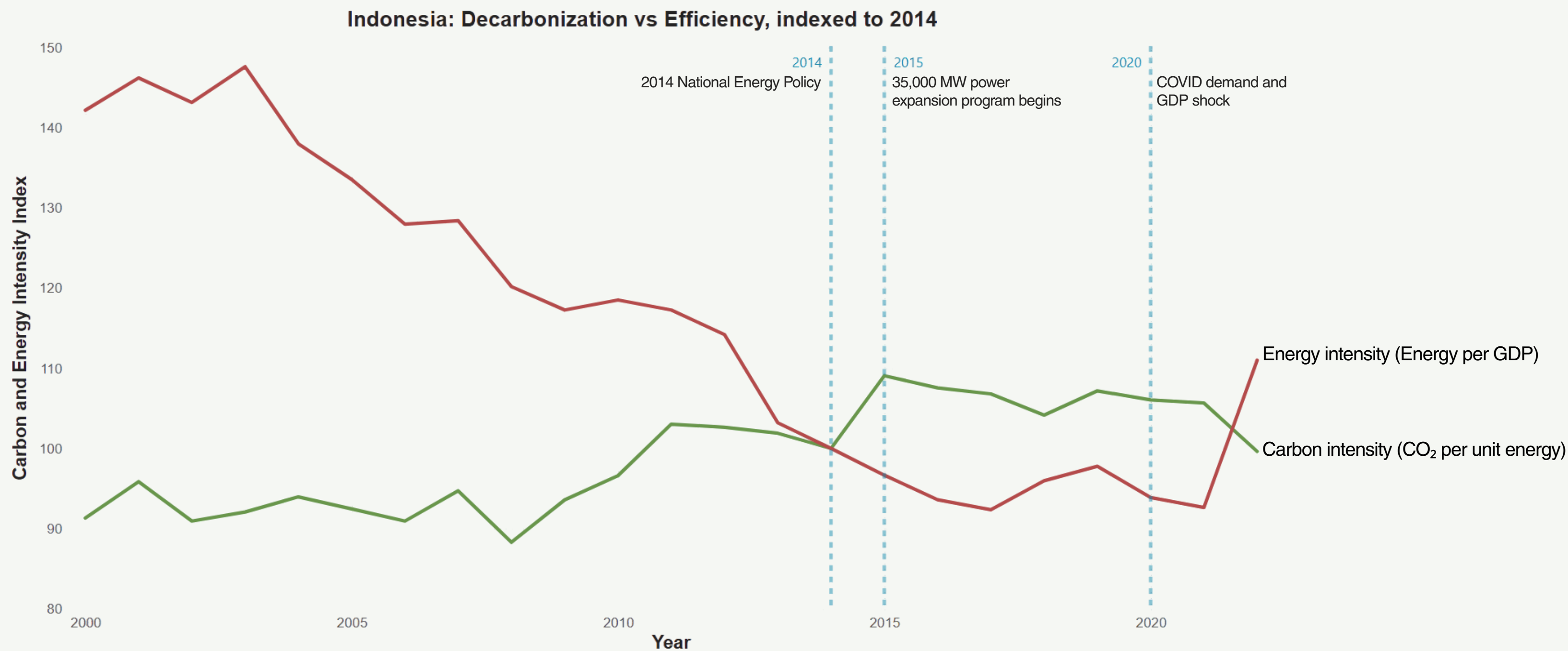


The Coal Trap

After 2014, efficiency improvements stalled (-3 Mt). Even worse, the grid became dirtier (+45 Mt) due to the coal boom.

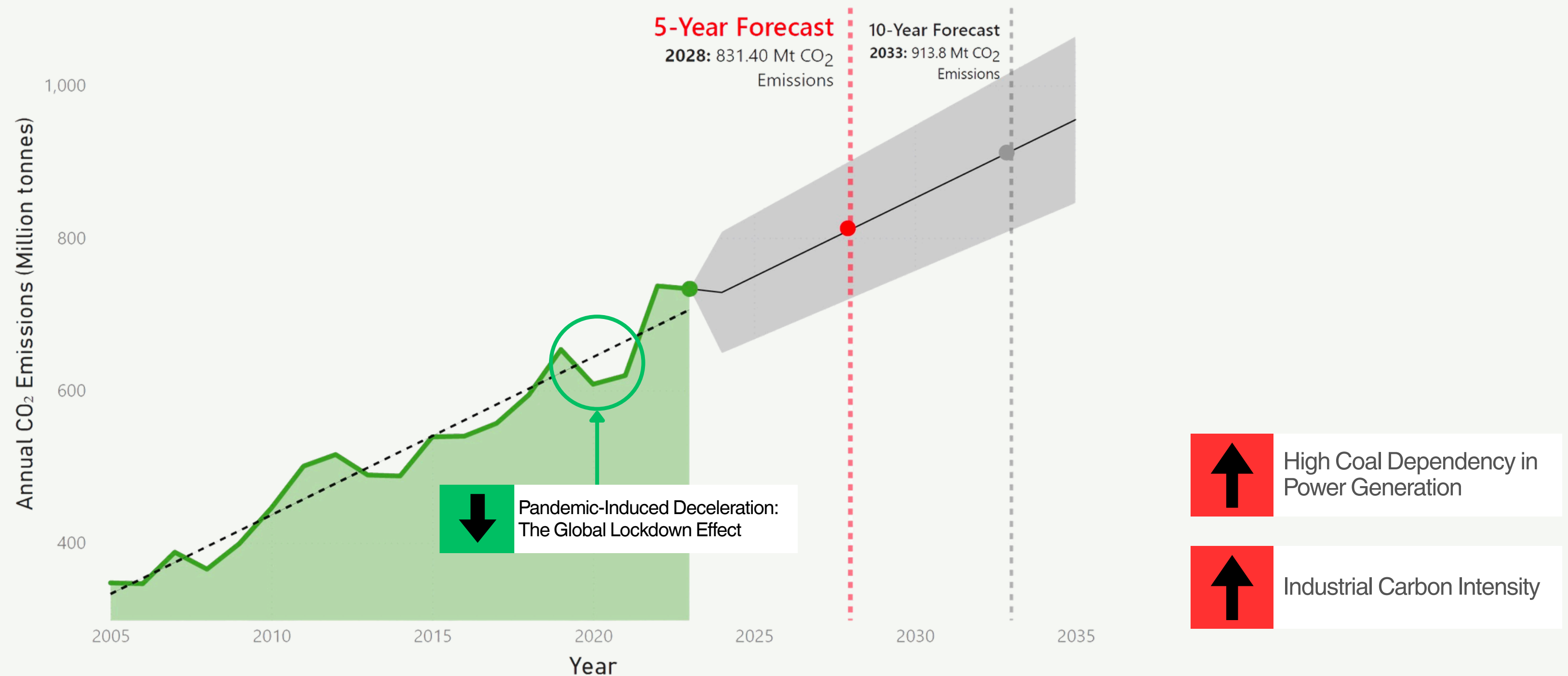
TRENDS IN ENERGY AND CARBON INTENSITY

Indexed to 2014 (2014 = 100). Efficiency improved, but carbon intensity stayed above the baseline for much of the post-2014 period.

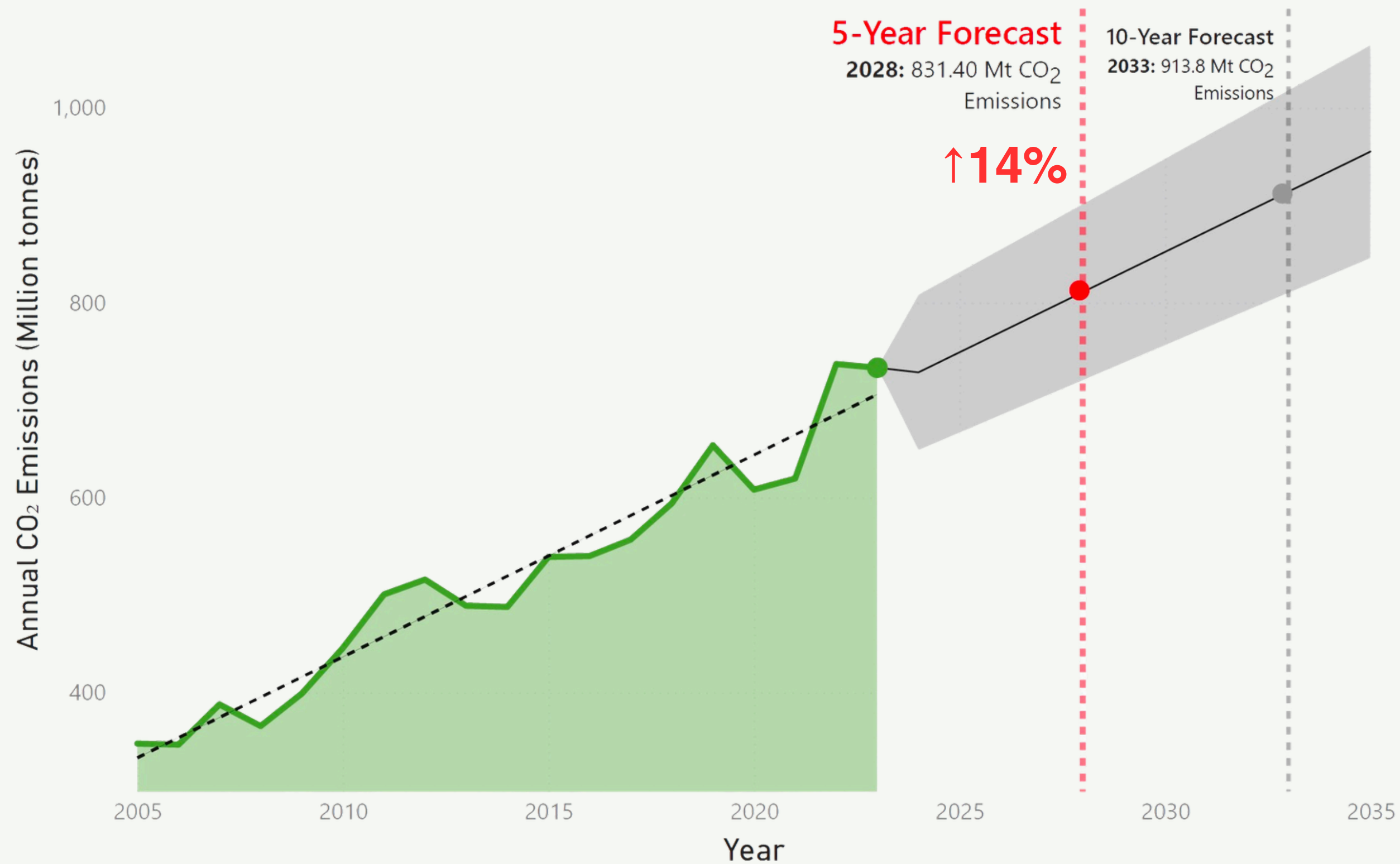


INDONESIA'S 5-YEAR CO₂ EMISSION FORECAST

A 5-Year Projection of Indonesia's CO₂ Emission

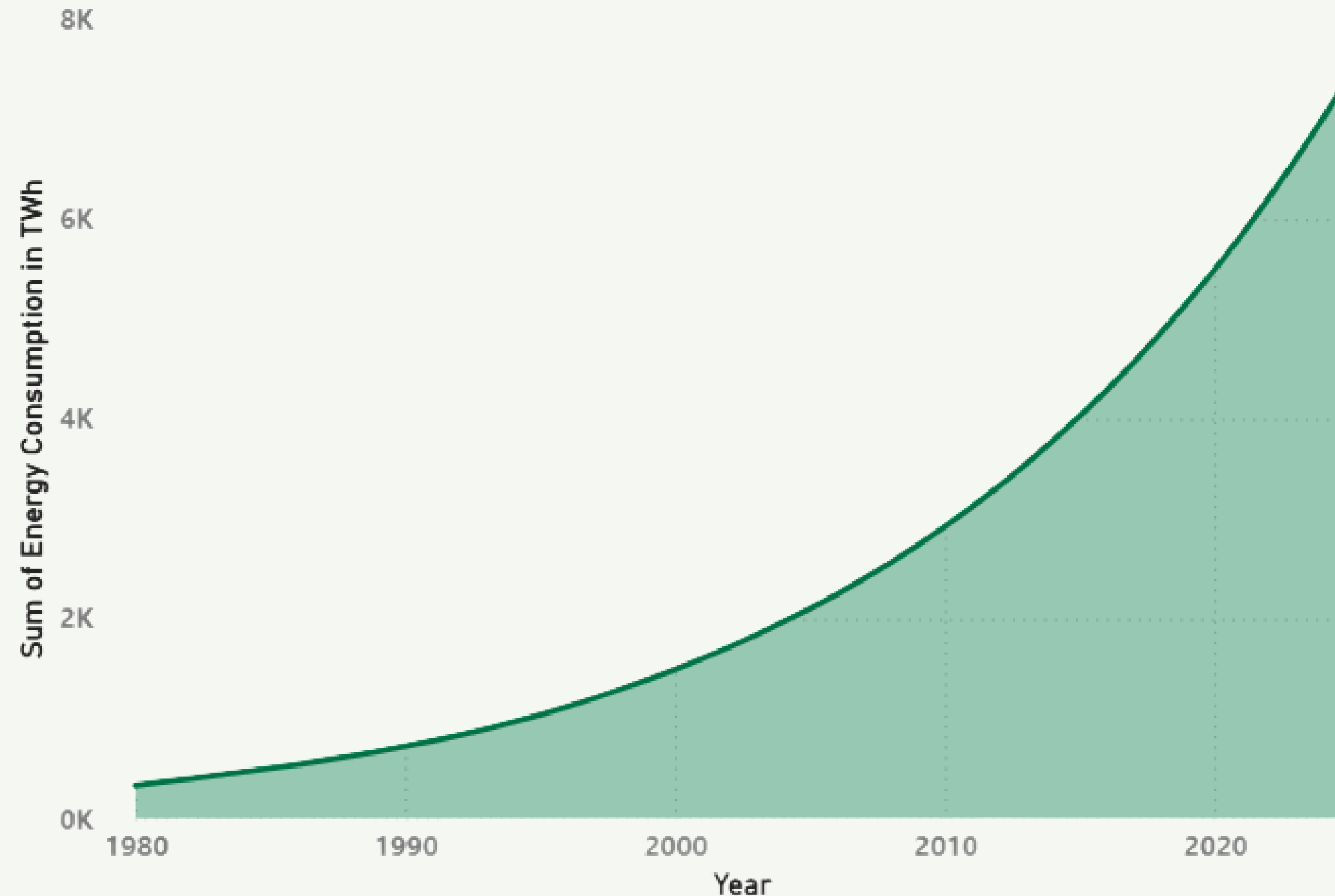


INDONESIA'S 5-YEAR CO₂ EMISSION FORECAST



ADDITIONAL DATA AND RECOMMENDATIONS

Indonesia's National Energy Demand Trends: 1980 - 2023

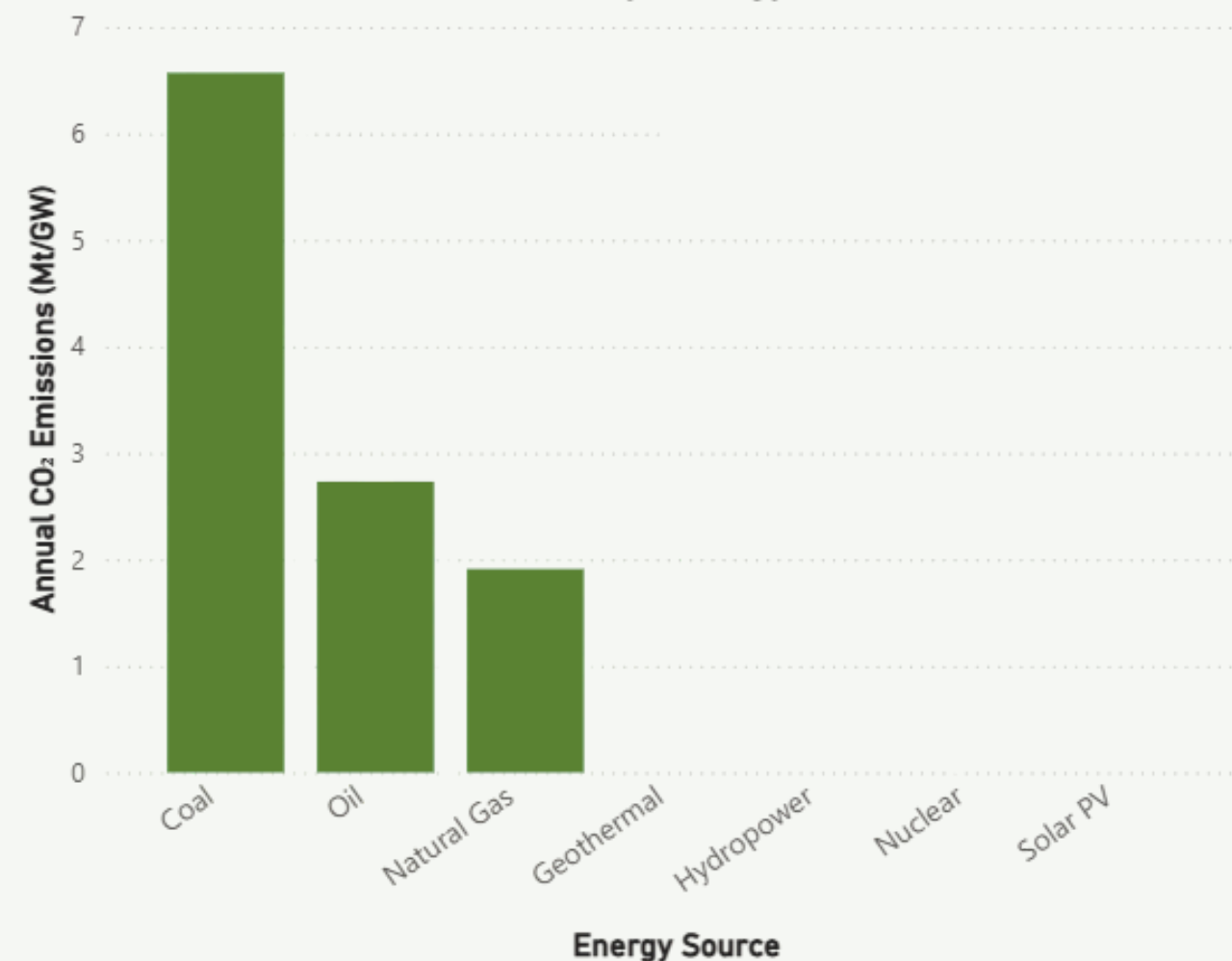


- There is a significant increase of energy consumption from 1980 to present
- There has also been a significant surge of consumption in the last decade.

ADDITIONAL DATA AND RECOMMENDATIONS

SUGGESTED Energy Sources According to CO2 Emmissions and Energy Density per Required Land Area

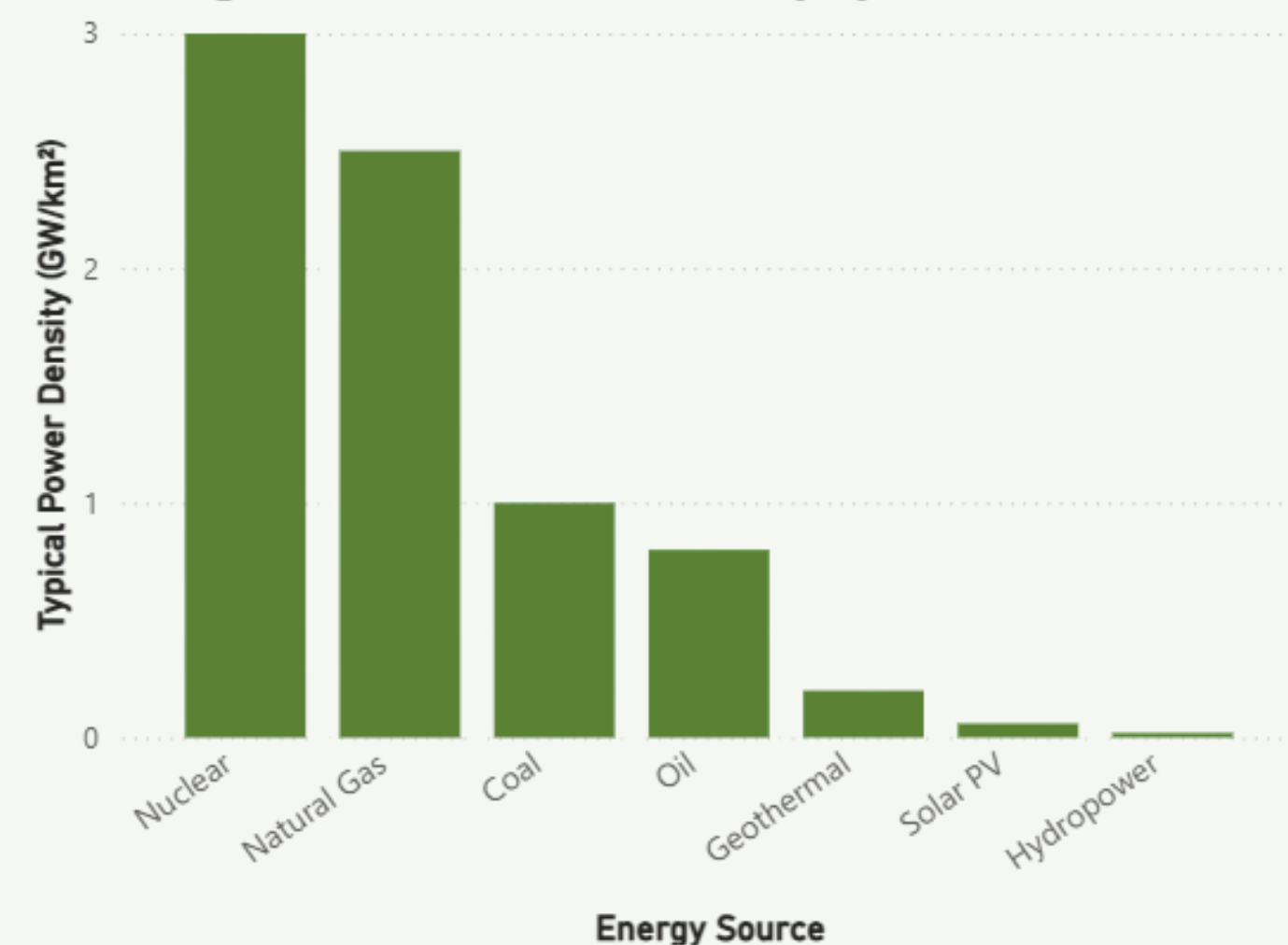
Annual CO₂ Emissions (Mt/GW) by Energy Source



Key Criteria for Selecting Additional Energy Sources

The primary factors to consider when choosing additional energy sources should include:

Average Power Generation Density by Source (GW/km²)



1. The lowest land requirement with the highest energy density.
2. The lowest CO₂ emissions paired with the highest energy density.

Based on these criteria, both nuclear and geothermal sources emerge as the most viable options.

ADDITIONAL DATA AND RECOMMENDATIONS

Theoretical Recommendations Based on Chosen Criteria

Energy Source	Energy to Land Area Ratio (GW/km ²)	Land Needed (km ²)	Added Energy Supply (GW)	Annual CO ₂ Reduction (Mt)
Nuclear	2.00	1.50	3.00	23.64
Geothermal	0.13	8.00	1.00	7.01
Total		9.50	4.00	30.65

Energy Plant Proposals and Emission Reductions

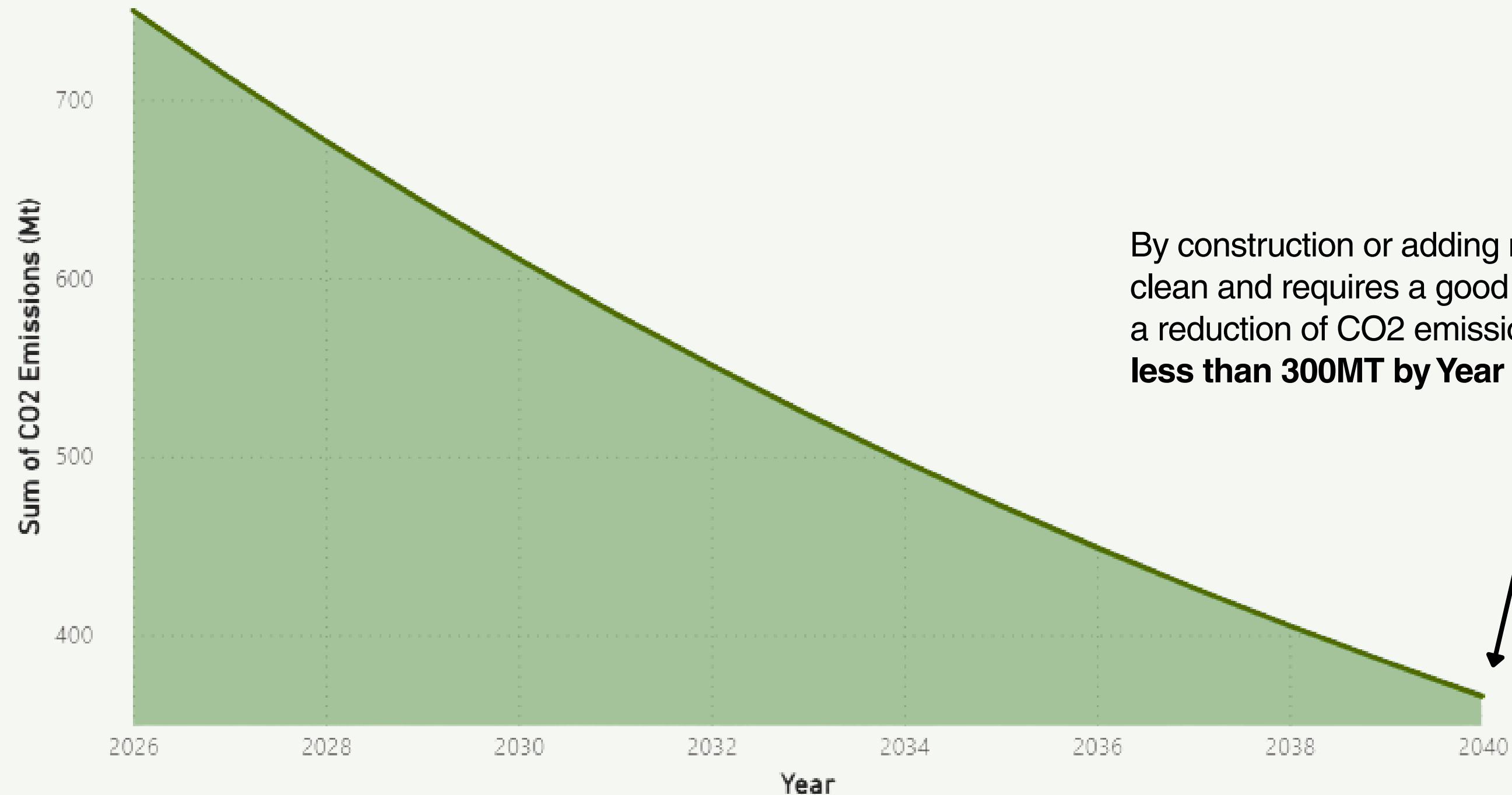
The implementation of these two energy plants could lead to an overall **5% reduction** in CO2 emissions **annually**.

1. Constructing a Nuclear Power Plant on **1.5 km²** can generate an additional **3 GW** of energy and decrease CO2 emissions by **23.64 Mt/year**.
2. Establishing a Geothermal Energy Plant over **8 km²** can produce an extra **1 GW** of energy and cut CO2 emissions by **7.01 Mt/year**.

ADDITIONAL DATA AND RECOMMENDATIONS

Graphical Presentation of CO₂ Decrease of 5% Per Year

Indonesia's Projected Annual CO₂ Emissions Reduction, 2026–2040



By construction or adding new energy sources that are clean and requires a good GW:Km² ratio, we can forecast a reduction of CO₂ emissions **from 750MT in 2026 to less than 300MT by Year 2040.**



CONCLUSION

Indonesia's emissions trajectory highlights the need for structural energy transitions to reverse long-term growth.

Key Insights

- Emissions growth is structural, not temporary
- Coal-based power and industry dominate recent increases
- Economic growth is the main driver for carbon emissions
- Current trends point to continued emissions growth

Strategic Direction

- Prioritize high-output, low-emission energy sources
- Reduce reliance on coal-dependent baseload and captive plants
- Align emissions reduction with long-term energy demand

